

variants either duplicated examples or made their prompt–response templates more regular. Exact duplication is deliberately content-free, as it increases the nominal number of stored examples without adding a new training case. But under epoch-based training, it also changes optimization exposure and revisiting frequency.

Though they carried a lot of negative results and imprecisions, these initial pilots let us rule out a lot of easy explanations: nominal rank is not a rate, nominal example count is confounded with training exposure, superficial corpus compressibility is not a clean information intervention, and a small exact-accuracy benchmark is too noisy. This led to the next question: what happens if we hold one learned update fixed and vary only how many bits are used to store it?

3.3 Fixed-checkpoint exact-rate protocol